

Traceable Reproduction Audit and Feedback-Assisted Diagnostics for VLMPC-Style Language-Table Pushing

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Abstract—Vision-language model predictive control (VLMPC) links semantic perception with receding-horizon control, but a reproduction study must distinguish an executable implementation from reliable closed-loop behavior. This paper audits a VLMPC-style Language-Table pushing system under a traceable protocol covering multi-seed controller reruns, language and image inputs, parameter ablations, event-triggered VLM re-query, target-image grounding, and visual-feedback boundaries. A fresh four-variant raw-MPC sweep over seeds 45–48 contains 16 runs and obtains 0/16 successes at the $d_{\text{world}}^{\text{final}} \leq 0.08$ evaluation tolerance; the archived eight-run parameter ablation also remains at 0/8. Feedback-assisted diagnostics use privileged simulator-state feedback: grounded original MPC obtains 4/4 and 8/8 diagnostic successes, with 23/23 raw planner actions replaced in all 12 runs, while the latest semantic-MPC suite obtains 18/18 selected-target control successes with mean final distance 0.0641 ± 0.0150 (sample SD). Only 17/18 rows preserve the requested-to-selected target identity, so this is not 18/18 end-to-end instruction success; the control result is also threshold-dependent, falling to 3/18 at 0.05. A fresh target-image matrix covers three cases and seeds 42–44: semantic selection and joint active control each succeed in 8/9 trials, including one retained yellow-pentagon selection failure. Fresh matched event controls show that natural re-query executes in 3/3 enabled trials but leaves final distances identical to no-event controls; a separate forced semantic-fault audit repairs the target in 3/3 trials and reaches the true-target threshold in 2/3. A detector/tracker visual-feedback implementation is also tested and fails in 0/6 trials, with mean final distance 0.3160 ± 0.1439 . The resulting contribution is a negative raw-reproduction audit plus bounded diagnostic evidence, not a benchmark-level proof of unmodified VLMPC success.

Keywords—Vision-language model predictive control, semantic MPC, Language-Table, reproduction audit, event re-query, target-image grounding.

I. INTRODUCTION

Robotic manipulation systems increasingly need to connect semantic understanding, visual perception, and closed-loop control. A table-top pushing task appears simple, yet a successful controller must recognize the relevant object, ground the goal, choose a feasible push direction, recover from perception drift, and stop near the target region. Model predictive control (MPC) is useful in such settings because it repeatedly predicts future states, optimizes candidate controls, and executes only

the first action before replanning [1], [2]. Classical MPC, however, usually assumes that state variables and objectives are already available. Vision-language models (VLMs) and vision-language-action systems add semantic grounding and instruction understanding [3]–[5], but they still require reliable low-level control to produce stable physical behavior.

VLMPC aims to bridge these two sides by placing VLM-based semantic evaluation inside a visual MPC loop [6]. The original framework takes a goal image or language instruction, samples candidate action sequences, predicts future visual observations, and evaluates the predictions with a hierarchical cost. Later trajectory-level variants further suggest that reducing the burden of video prediction can improve long-horizon manipulation planning [7]. For a constrained local reproduction study, the key question is therefore not only whether the program can run, but what experimental evidence supports each success claim.

This paper summarizes our study around that question. We use Language-Table push-to-corner tasks and evaluate the unmodified video-prediction MPC chain, grounded and semantic oracle-feedback diagnostics, a target-image interface, event-triggered VLM re-query, and a detector/tracker visual-feedback boundary test. The central reporting choice is to keep negative results visible. The fresh unmodified sweep is executable and logged, but remains unsuccessful across all 16 controller-variant/seed combinations. The positive 4/4, 8/8, and selected-target 18/18 numbers come from feedback-assisted diagnostic paths and are therefore reported with an explicit state-source and target-reference flag; the semantic suite contains 17/18 requested-to-selected target matches. In contrast, the visual-feedback implementation is tested without simulator-state access and also fails in 0/6 trials. This separation prevents an oracle-stabilized engineering path from being mistaken for a faithful reproduction of the original video-prediction controller or for a solved visual-feedback system.

The contributions are:

- 1) A traceable reproduction audit of a VLMPC-style Language-Table manipulation chain, including unmodified controller comparison and original parameter ablation.

- 2) A conservative symptom analysis that connects failed runs to logged tracking jumps, visual-prediction/control mismatch, parameter sensitivity, and absence of final-action feedback, without claiming single-factor causality.
- 3) Two bounded diagnostic enhancements: grounded original MPC, which preserves raw planner calls but applies final action correction, and semantic MPC, which combines semantic target selection with state-feedback pushing.
- 4) A branch-level accounting of planner authority, including the grounded branch’s raw-action replacement rate and the semantic branch’s zero active raw-sampling budget.
- 5) A file-traceable experimental package with sanitized per-run and aggregate CSVs, publication figures, qualitative frames, matched event controls, a repeated target-image end-to-end matrix, a visual-feedback boundary test, and a controlled semantic-fault re-query audit.

II. RELATED WORK

A. Visual MPC and Video-Prediction Control

Visual MPC uses learned visual dynamics to support model-based manipulation directly in image space. Deep visual foresight and related visual MPC systems showed that action-conditioned video prediction can guide robot motion without a hand-designed dynamics model [8], [9]. Such approaches are attractive for open-ended manipulation, but they are also sensitive to prediction bias, object occlusion, and long-horizon compounding errors. These limitations are relevant to our reproduction: the planner can generate candidate futures and rank actions, but the unmodified closed loop does not reliably drive the object to the selected goal region.

B. Vision-Language Manipulation

Language-conditioned manipulation has used VLMs and language models for task grounding, affordance selection, reasoning, and failure interpretation. CLIPort separates semantic “what” reasoning from spatial “where” execution [10]. SayCan grounds language plans with affordances [3]. Inner Monologue emphasizes feedback-aware embodied reasoning [5], and VoxPoser composes 3D value maps for robotic manipulation from language-model reasoning [11]. These systems motivate our use of explicit feedback and event handling: semantic decisions must be connected to low-level execution evidence, not only to high-level task labels.

C. VLMPC, Trajectory Variants, and Testing

VLMPC integrates VLM perception with MPC-style candidate action evaluation [6]. Traj-VLMPC further reduces computational overhead by using trajectory generation rather than full video prediction [7]. At the same time, evaluation work such as VLATest argues that VLM/VLA manipulation systems require robustness and reliability tests beyond isolated demonstrations [12]. Failure-reasoning systems such as AHA also show the value of identifying manipulation failure modes

[13]. Our work follows this evaluation-oriented line: we do not hide failed unmodified runs, and we mark diagnostic controllers that use privileged state feedback.

III. EVALUATION SETUP AND EVIDENCE SOURCES

A. Task Definition

The experiment uses Language-Table desktop pushing tasks. A target object such as *red moon*, *blue cube*, or *yellow pentagon* should be pushed toward a designated corner. The controller input can be a fixed target label, a natural-language instruction, a target image, or a VLM-based scene selection request. The action is a planar end-effector push in the simulated table environment.

The primary metric is the final world-coordinate distance between the target object and the goal:

$$\text{success} = \mathbb{I} [d_{\text{world}}^{\text{final}} \leq 0.08]. \quad (1)$$

The threshold 0.08 is an evaluation tolerance used consistently in the archived final runs, not a universal manipulation success criterion. Threshold sensitivity changes the interpretation of the positive results: at 0.05, the archived grounded diagnostic suites drop to 0/4 and 0/8, and the latest semantic MPC suite succeeds in only 3/18 runs. We therefore report success rates primarily as evidence of coarse task completion under the evaluation tolerance, not precision placement.

For the live semantic suite, the distance-based control label is computed relative to the selected target recorded for that diagnostic condition. We additionally compare the requested and selected target names: 17/18 rows match, and the corresponding instruction-level joint count is 17/18. Thus the reported 18/18 value is a conditional selected-target control result, not an end-to-end language-instruction success rate.

For the new target-image matrix, we distinguish a *pre-satisfied* run, for which the true target is already within the threshold before the first action, from an *active control* trial, which starts outside the threshold and executes at least one action. *Joint active success* requires both correct target-image semantic selection and active control success. This prevents a zero-step terminal state from being counted as controller evidence.

B. Evidence Artifacts

All numerical results in this paper are extracted from the study package. The main evidence files are:

- sanitized per-run and aggregate CSVs for the fresh raw-MPC sweep, semantic suite, target-image matrix, event controls, semantic-fault audit, and visual-feedback test;
- supplement CSVs for exact method parameters, checkpoint/code hashes, target-image selection, and VLM cache status;
- publication figures and selected qualitative frames used as visual evidence; raw service logs and videos remain outside the sanitized public package;
- a portable robustness-edge audit CSV covering occlusion, viewpoint proxy, brightness shift, and unknown-target rejection, together with its four input images.

The reporting goal is to make every claim traceable to these artifacts rather than to a post-hoc manual description. The online semantic calls use a GPT-5.5-based VLM through an external Responses-compatible runtime/interface; the model alias is reported for reproducibility, while credentials and the private endpoint are excluded from the manuscript and sanitized package. The revision controls use strict no-cache mode and record zero cache hits. They still do not prove that natural re-query causes recovery because matched no-event controls also succeed. The public repository is available at <https://github.com/yixiaogithub66/vlmpc-language-table-reproduction-audit>; the accompanying v1.0.6 revision artifact contains the authoritative result files, including the robustness-edge CSV and portable inputs, reproduction guide, parameter records, publication-figure source and exports, separate runtime-asset and source-snapshot manifests, and a reviewer-facing source snapshot. Third-party model weights are not redistributed and must be obtained from their original sources.

IV. METHOD

A. Unmodified Reproduction Chain

The unmodified reproduction follows a VLMPC-style loop. A task input is grounded to a target, candidate action sequences are sampled, a visual prediction model forecasts future observations, and a cost function ranks the candidates before the first action is executed. Controller comparison includes baseline, improved, literature, and literature-v2 settings. Parameter ablation varies sampling budget and cost settings.

In this branch, no final-action state-feedback correction is applied. This distinction is crucial: these runs represent the closest archived reproduction of the original video-prediction MPC chain in our project, and they are evaluated separately from later enhancements.

B. Archived Implementation Parameters

Table I summarizes the recoverable implementation parameters. The unmodified planner uses horizon $H = 20$, action dimension 2, and a correlated-noise sampler with $\beta = 0.5$. The command-line defaults are 200 samples and replan frequency 5, while the archived ablation set contains eight logged configurations rather than a complete 3×3 grid. The literature-style cost is

$$J = d_{\text{obj},\text{goal}} + 0.35d_{\text{contact}} + 0.15d_{\text{arm,obj}} + 18.0J_{\text{align}} + 24.0J_{\text{obs}} - 1.2\Delta d + 12.0\mathbb{I}[\Delta d < 1.0], \quad (2)$$

with obstacle decay 42 pixels. In the close-range branch, the contact, alignment, progress-reward, and non-progress terms are changed to 0.10, 6.0, 1.8, and 20.0, respectively. These values are implementation parameters recovered from `vlmpc.py`, not tuned benchmark claims.

The runtime-asset manifest records SHA-256 prefixes for the externally obtained video-prediction, detector, and tracker assets. The source manifest separately records the executed-source and public-snapshot hashes. Five listed Python files

TABLE I
RECOVERABLE IMPLEMENTATION PARAMETERS FROM CODE, METRICS,
AND SUPPLEMENT CSVs.

Parameter	Value or rule used in the combined audit
Environment	BLOCK_8 Language-Table, 10 Hz; fresh raw sweep uses seeds 45–48; semantic, target-image, and event controls use 42–44; visual feedback uses 45–50
Raw MPC	action dim. 2, horizon 20; sampled candidates clipped to $[-0.02, 0.02]$, executed action/feedback bound $[-0.08, 0.08]$
Budgets	default 200/5 samples/plan frequency; eight archived ablations: pf1/ns5, pf2/ns5, pf2/ns10, pf3/ns10, pf2/ns15, cost history low, cost default, cost target high
Detector/tracker	YOLO conf. 0.5; Pysot SiamRPN tracker
Cost weights	contact/approach/alignment/obstacle = 0.35/0.15/18.0/24.0
Progress terms	reward 1.2, non-progress penalty 12.0, min gain 1.0, obstacle decay 42 px
Close range	distance 135 px; contact/alignment/reward/penalty = 0.10/6.0/1.8/20.0
Feedback push	contact offset 0.06, clearance 0.12, move step 0.06, push step 0.075, tolerance 0.018
Original event	default window/min-progress/min-step = 3/1.0/3; matched controls use 2/25.0/3
Semantic event	window 4, min progress 0.002, distance floor $1.5 \times$ success distance
Online VLM	GPT-5.5-based VLM through an external Responses-compatible runtime/interface; credentials and endpoint excluded
Success threshold	evaluation tolerance $d_{\text{world}} \leq 0.08$; strict audit also reports 0.05 and 0.10

are byte-identical. For `main.py`, the executed-source prefix is `d8078e7cba86` and the public-snapshot prefix is `f98caece7442`; the difference is limited to replacing a drive-local dependency fallback with environment-based configuration, without changing controller logic.

C. Grounded Original MPC

Grounded original MPC is a diagnostic enhancement. The raw planner path still runs, and raw MPC actions are recorded. After the raw action is obtained, a simulator-state feedback push correction is used for the executed action. The correction is bounded by action clipping and uses object, end-effector, and goal positions available from the simulator state. Across the 12 grounded runs (4 controller-comparison runs and 8 ablation runs), the logs contain 23 raw planner actions and 23 feedback overrides per run: the executed-action replacement rate is therefore 23/23, or 100%, in every run. The raw planner remains useful for traceability and branch comparison, but it is not the authority over the executed action in this diagnostic branch.

The grounded and semantic branches use simulator-state feedback and therefore answer a diagnostic question rather than a reproduction question. The feedback action uses object, end-effector, and goal positions from simulator state. Let $\hat{d}_{\text{push}} = (x_{\text{goal}} - x_{\text{obj}}) / \|x_{\text{goal}} - x_{\text{obj}}\|$ and $x_{\text{contact}} = x_{\text{obj}} - 0.06\hat{d}_{\text{push}}$. The controller first stages the end effector near a side-clearance point, then moves to contact, and finally pushes:

$$u_t = \text{clip} \left(\alpha \hat{d}_{\text{push}} + \beta (x_{\text{contact}} - x_{\text{ee}}), -0.08, 0.08 \right). \quad (3)$$

In the push phase, $\alpha = 0.075$ and $\beta = 0.45$; the stage/contact phases instead use a step-limited move of at most 0.06.

D. Semantic MPC

Semantic MPC is a second diagnostic path. It uses semantic target selection from fixed labels, language instructions, target images, or VLM scene selection, and then performs low-level pushing with state feedback. The strict repair adds an `auto|vlm|vision` target-image selector: in `auto` mode the system sends the target image and the current scene image to the configured VLM and asks it to choose among visible scene candidates. The outer loop retains repeated receding-horizon execution and logging, but the final semantic push uses no active raw video-prediction sampling budget ($N_{\text{raw}} = 0$); the state-feedback law, rather than a cost-ranked raw action, determines the executed push. It is therefore a stabilized semantic-control baseline within this project, not a faithful reproduction of the raw video-prediction MPC chain.

E. Event-Triggered VLM Re-query

The event mechanism is designed to avoid repeated VLM calls unless tracking or progress conditions become unreliable. In the fresh matched controls, the natural re-query is not forced: `forced_initial_requery` is false for the no-event and event-enabled rows. The 2026-09-22 controls run with cache disabled, record zero cache hits, and log fresh responses through the Responses-compatible interface. Natural event re-query fires in all three event-enabled rows, but no-event and event-enabled controls both succeed with identical final distances for each seed. The data therefore support an implementation claim that the event path can fire, but not a causal recovery claim.

To make the recovery question measurable without relabeling a naturally successful run, the revision adds a separate paired semantic-fault audit. Each seed starts with the true target replaced by a deliberately wrong initial target (blue cube instead of red moon). The treatment forces one explicitly labeled synthetic audit re-query at step 3; the control keeps the wrong target for the run. Evaluation is performed against the independent true target, and synthetic re-queries are reported separately from natural detector events. This audit tests whether the implementation can repair a known semantic corruption; it does not estimate natural-event sensitivity or establish a general causal benefit.

V. EXPERIMENTS

A. Experiment Groups

We organize experiments into three evidence groups. **Raw reproduction** is the fresh four-variant, four-seed sweep of the video-prediction MPC chain without final-action state feedback; the earlier eight-run parameter ablation is retained as an explicitly archived comparison. **Oracle-feedback diagnostics** contains grounded execution, semantic MPC, target-image grounding, and event controls that use privileged simulator state for low-level execution. **Visual-feedback boundary** replaces the simulator-state accessor with the implemented detector/tracker path and measures the resulting boundary on six seeds. These groups are not pooled as if they were the same controller.

B. Unmodified Reproduction Results

Table III shows the fresh raw-MPC sweep. Every controller variant fails on every seed: 0/16 at the project threshold, with variant means from 0.3134 to 0.3358 and an aggregate mean of 0.3245. The failures remain 0/16 at both 0.05 and 0.10. The archived eight-run parameter ablation also remains 0/8, but it is not counted as an independent extension of the fresh seed sweep. This is the main negative result of the project and should remain visible in any final presentation.

C. Feedback-Assisted Diagnostic Results

Table IV reports the diagnostic branches. The privileged-state column changes the interpretation of the positive results. Grounded original MPC reaches 4/4 and 8/8, but all 12 runs replace 23/23 raw planner actions, so the feedback law rather than the raw planner has execution authority. The latest semantic-MPC suite reaches 18/18 only when success is defined relative to the selected target; 17/18 rows preserve the requested-to-selected target identity and therefore 17/18 are instruction-level joint successes. Final selected-target distances range from 0.0297 to 0.0798; at 0.05 the selected-target control count is only 3/18. The fresh target-image matrix has no pre-satisfied cases: semantic selection and joint active control each succeed in 8/9 trials, with the yellow-pentagon seed-42 error retained. The detector/tracker visual-feedback path is a separate non-oracle test and fails in 0/6 trials. The paired semantic-fault audit is reported as a controlled repair experiment, not as evidence that natural events improve ordinary runs.

D. Per-Run Evidence Digest

Table VI gives a compact digest extracted from the portable CSVs. The full files retain seeds, labels, final distance, steps, runtime, request counts, selector metadata, and sanitized provenance. The table highlights three facts: every fresh raw-MPC run remains outside the threshold; oracle-feedback diagnostics succeed only within an explicitly privileged branch; and the first detector/tracker feedback implementation does not close that gap.

E. Event Re-query Evidence

The event evidence is reported as an implementation and control supplement. The fresh 2026-09-22 set contains nine rows: three matched no-event controls, three matched event-enabled controls, and three forced scene-selection audits, all at seeds 42–44. Natural event re-query fires in 3/3 event-enabled rows; the no-event and event-enabled final distances are identical seed by seed. All rows use strict no-cache mode and record zero cache hits. These controls verify trigger execution but do not prove that natural re-query caused recovery. The separate semantic-fault audit below is deliberately evaluated against an independent true target and is reported as a controlled repair test.

The treatment reaches the correct semantic target in all three paired runs, but one paired run ends just above the 0.08 threshold. The result supports a narrower statement: an explicitly forced re-query can repair a known semantic

TABLE II

CONTROLLER-BRANCH RESPONSIBILITIES AND ACTION AUTHORITY. “PRIVILEGED” DENOTES DIRECT SIMULATOR-STATE ACCESS DURING EXECUTION.

Branch	Semantic/planning role	Actual MPC role	Executed action authority	State access
Unmodified VLMPC	Ground target; sample action sequences; predict visual futures	Candidate ranking with a video-prediction cost; first raw action is executed	Raw planner action	Visual observations only
Grounded original MPC	Preserve the same raw planner call and log its actions	Raw planner remains in the loop for comparison and traceability	Privileged state-feedback correction replaces the raw action (23/23 in all 12 runs)	Simulator state
Semantic MPC	Select target from label, language, image, or VLM scene query	Receding-horizon execution/logging; no active raw video-prediction sampling ($N_{\text{raw}} = 0$)	Privileged state-feedback push law	Simulator state
Visual feedback	Select a fixed target; estimate control state from RGB observations	Receding-horizon detector/tracker feedback; no simulator-state accessor	Visual detector/tracker feedback law	RGB observations
Event-control supplement	Monitor stagnation/semantic events and refresh the VLM instruction when triggered	Event logic controls query timing, not the low-level push optimizer	Same semantic-MPC feedback action	Simulator state

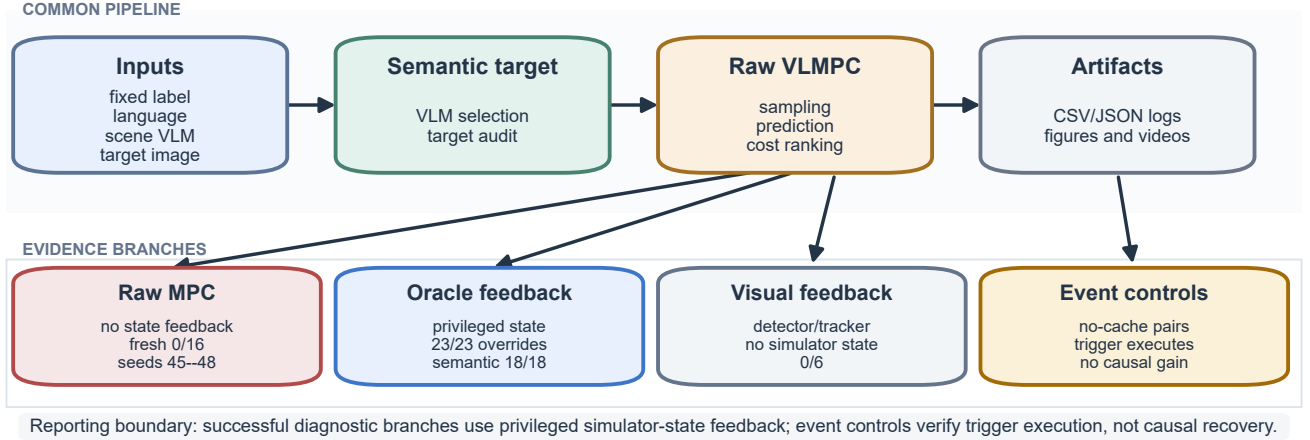


Fig. 1. Strict method overview. The common pipeline is separated from the evidence branches. Grounded original MPC and semantic MPC use privileged state feedback for execution, so their successes are reported as diagnostic stabilization rather than unmodified VLMPC success. The event-control path verifies trigger execution under matched no-cache controls, not causal recovery.

TABLE III

FRESH RAW-MPC REPRODUCTION SWEEP. NO PRIVILEGED STATE FEEDBACK IS USED; FINAL DISTANCE IS MEAN $\pm$ SAMPLE SD OVER SEEDS 45–48. THE ARCHIVED ABLATION ROW IS SHOWN SEPARATELY AND IS NOT POOLED WITH THE FRESH SWEEP.

Variant/suite	Runs	Succ.	Final dist.	Range
Baseline	4	0/4	0.3358 ± 0.1099	0.199–0.436
Improved	4	0/4	0.3354 ± 0.1048	0.224–0.436
Literature	4	0/4	0.3134 ± 0.1325	0.163–0.436
Literature-v2	4	0/4	0.3134 ± 0.1325	0.163–0.436
All fresh raw runs	16	0/16	0.3245 ± 0.1085	0.163–0.436
Archived ablation	8	0/8	0.2534 ± 0.0304	0.216–0.294

Threshold sensitivity at 0.05/0.08/0.10 is 0/16, 0/16, 0/16 for the fresh raw sweep and 0/8, 0/8, 0/8 for the archived ablation.

fault in this setup. It does not establish that the natural stagnation detector finds all such faults, nor that natural query improves ordinary successful runs.

F. Target-Image Selection Audit

The target-image interface is separated into three evidence levels. The old detector/corner heuristic was fragile: it selected *blue cube* in three of the four live audit cases, giving only 1/4 full-frame heuristic matches. The repaired VLM semantic selector sends both the target image and the current scene image to the configured VLM, asks for a visible scene object, and records the raw response; it matches the expected object in

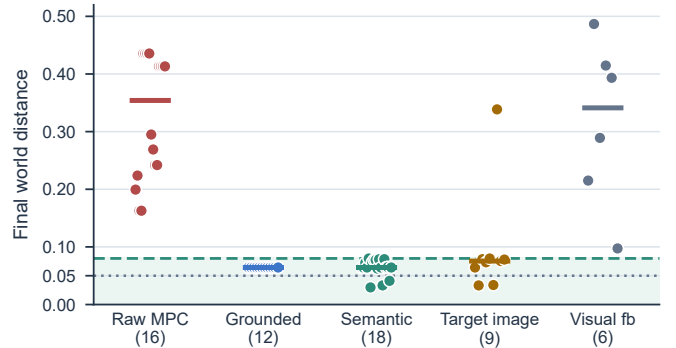


Fig. 2. Final-distance distributions across the fresh raw-MPC sweep, archived grounded diagnostics, live semantic and target-image suites, and the visual-feedback test. Each dot is one run and the horizontal segment is the within-suite median.

4/4 audited cases. The nine-run end-to-end matrix covers three cases and seeds 42–44, with 8/9 semantic matches and 8/9 joint active successes. The four-case audit remains semantic-selection evidence, while the matrix is still a small diagnostic control study rather than a broad target-image robustness benchmark.

TABLE IV

DIAGNOSTIC RESULTS AND FEEDBACK SOURCE. ORACLE ROWS USE PRIVILEGED SIMULATOR-STATE FEEDBACK AND ARE NOT DIRECT UNMODIFIED VLMPC REPRODUCTIONS; THE VISUAL-FEEDBACK ROW USES THE IMPLEMENTED DETECTOR/TRACKER PATH WITHOUT SIMULATOR-STATE ACCESS.

Suite	Runs	Control succ.	Evidence and semantic check	Final dist.	State
Grounded controller	4	4/4	target fixed; 23/23 override	0.0643±0.0000	yes
Grounded ablation	8	8/8	target fixed; 23/23 override	0.0643±0.0000	yes
Event controls	9	9/9	matched controls; natural trigger 3/3 enabled, 0/3 no-event	0.0587±0.0233	matched yes
Semantic MPC live	18	18/18*	17/18 request-selection matches; selected-target control	0.0641±0.0150	yes
Target-image matrix	9	8/9 joint	semantic 8/9; no pre-satisfied	0.0950±0.0932	yes
Visual feedback	6	0/6	detector/tracker path	0.3160±0.1439	no
Semantic-fault audit	6 paired	2/3 treatment	synthetic repair 3/3; control 0/3	0.0650±0.0182	treatment yes

* The semantic-MPC control count is conditional on the selected target. The portable CSV records 17/18 requested-to-selected target matches and 17/18 instruction-level joint successes; it is not an 18/18 end-to-end instruction result.

TABLE V

CONDITION-WISE VARIABILITY FOR THE LATEST SEMANTIC-MPC SUITE. VALUES ARE MEAN ± SAMPLE SD OF FINAL WORLD DISTANCE; THE 18 ROWS ARE HETEROGENEOUS DIAGNOSTIC RUNS, NOT 18 INDEPENDENT RANDOM-SEED REPETITIONS. THE SEED-ONLY SUBSET HAS $n = 3$.

Condition	n	Mean ± SD	≤ 0.05	≤ 0.08	Target/seed scope	Interpretation
Input forms	4	0.0663 ± 0.0041	0/4	4/4	3 forms + target image	Input-form comparison
Multi-target	8	0.0678 ± 0.0167	1/8	8/8	8 target objects	Small target sweep
Multi-seed	3	0.0587 ± 0.0233	1/3	3/3	red moon, seeds 42/43/44	Seed variability
Semantic ablation	3	0.0568 ± 0.0136	1/3	3/3	default/precise/fast	Semantic-control settings
All latest runs	18	0.0641 ± 0.0150	3/18	18/18	heterogeneous union	Run-level description only

TABLE VI

PER-RUN DIGEST EXTRACTED FROM ARCHIVED AND LATEST CSV FILES. EXEC. S IS DISTANCE-THRESHOLD SUCCESS UNDER $d_{\text{world}}^{\text{final}} \leq 0.08$.

Suite	Final-distance summary	Range	Exec. S	Semantic/VLM evidence
Fresh raw MPC	4 variants × 4 seeds; aggregate mean 0.3245 ± 0.1085	0.163–0.436	0/16	visual-prediction controller fails
Archived raw ablation	eight logged parameter settings	0.216–0.294	0/8	archived comparison
Grounded controller	all 0.064	0.064–0.064	4/4	fixed target
Grounded ablation	all 0.064	0.064–0.064	8/8	fixed target
Event controls	matched no-event/event distances identical by seed	0.033–0.079	9/9	natural trigger 3/3; no causal gain
Semantic MPC live	min 0.030; median 0.064; max 0.0798	0.030–0.0798	18/18	17/18 request-selection matches; selected-target control
Target-image matrix	8 semantic matches; one yellow-pentagon error	0.033–0.339	8/9 joint	all nine are active trials
Visual feedback	detector/tracker feedback, seeds 45–50	0.097–0.487	0/6	no simulator-state access
Semantic-fault revision audit	treatment repairs target in 3/3; control 0/3	0.045–0.385	2/3 treatment	synthetic re-query; natural count 0

TABLE VII

FRESH EVENT RE-QUERY CONTROLS FROM THE STRICT NO-CACHE 2026-09-22 EXPERIMENT.

Case	Natural re- query	Final dist.	Note
No-event controls, seeds 42–44	0	0.0332–0.0787	3/3 success; 3 fresh responses
Event-enabled controls, seeds 42–44	1 each	0.0332–0.0787	3/3 success; 6 fresh responses
Forced scene audit, seeds 42–44	1 natural + 1 forced	0.0345–0.0644	3/3 success; 18 fresh responses

TABLE VIII

CONTROLLED SEMANTIC-FAULT RE-QUERY AUDIT. THE WRONG INITIAL TARGET IS INJECTED DELIBERATELY; SUCCESS IS MEASURED AGAINST THE INDEPENDENT TRUE TARGET.

Condition	n	Semantic repair	True-target success	Final dist.
No re-query control	3	0/3	0/3	0.2950 ± 0.0868
Synthetic re-query	3	3/3	2/3	0.0650 ± 0.0182

Threshold sensitivity

Fresh raw MPC (n=16)	0/16	0/16	0/16
Archived ablation (n=8)	0/8	0/8	0/8
Grounded diagnostics (n=12)	0/12	12/12	12/12
Semantic MPC (n=18)	3/18	18/18	18/18
Target image (n=9)	2/9	8/9	8/9
Visual feedback (n=6)	0/6	0/6	1/6
	≤ 0.05	≤ 0.08	≤ 0.10

Fig. 3. Threshold sensitivity at 0.05, 0.08, and 0.10, shown as exact selected-target control counts. The outlined column is the project threshold. Fresh raw MPC fails in 0/16 at all three thresholds; semantic MPC changes from 3/18 at 0.05 to 18/18 at 0.08, while requested-to-selected matching is 17/18.

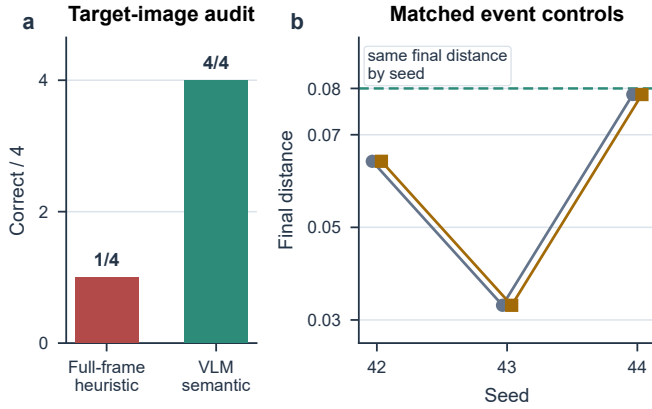


Fig. 4. Target-image and event-control audits. (a) The live VLM semantic selector is correct in all four audited cases, whereas the old full-frame heuristic is correct in only one. (b) Fresh strict no-cache matched controls have identical no-event and event-enabled final distances by seed; all nine rows record zero cache hits.

TABLE IX
TARGET-IMAGE SEMANTIC-SELECTION AUDIT. VLM SEMANTIC CHOICE USES THE PROJECT VLM BACKEND WITH TARGET-IMAGE AND SCENE-IMAGE INPUTS.

Case	Exp.	Old	VLM	OK
Red-moon crop	red moon	blue cube	red moon	yes
Blue-cube crop	blue cube	blue cube	blue cube	yes
Yellow-pent. crop	yellow pentagon	blue cube	yellow	yes
Archived frame	red moon	blue cube	pentagon	yes

G. Revision Target-Image End-to-End Matrix

The revision repeats the target-image path end to end for three manually specified crops and seeds 42–44. The VLM receives the target image and current scene image, selects a named scene object, and the selected target is then passed to the same semantic-MPC feedback controller. All nine trials begin outside the 0.08 threshold, so there is no pre-satisfied denominator adjustment. The yellow-pentagon run at seed 42 selects *yellow star*, remains at distance 0.3386, and is retained as the negative case.

The complete per-run table is included in the revision target-image CSV under the sanitized supplement. This expanded matrix strengthens the end-to-end claim relative to the earlier one-run path, while the incorrect yellow-pentagon selection keeps the result bounded.

H. Observed Failure Symptoms

The failure analysis is reported as symptom taxonomy rather than causal proof. The fresh four-seed sweep fails for every raw controller variant, and the archived ablation also fails, indicating that the tested variants and parameter changes do not solve the reproduced control problem. The detector/tracker feedback path additionally fails in all six visual-feedback trials. Table XI summarizes the evidence and interpretation.

I. Robustness-Boundary Checks

The robustness-edge audit is not a large benchmark, but it provides boundary evidence for the perception and target-

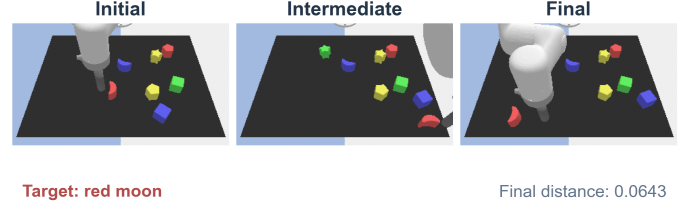


Fig. 5. Representative frames from an oracle-feedback red-moon execution. The quantitative claims are taken from the portable run tables; the frames provide qualitative evidence of a logged Language-Table execution rather than a table-only report.

mapping layer. Its portable CSV and four input images are included in the release artifact. The detector still identifies the red moon under target occlusion, viewpoint proxy, and brightness shift, with confidence decreasing under occlusion. Unknown or unsupported target names are safely rejected rather than mapped to arbitrary known objects. These are detector-level observations, not closed-loop robustness or real-camera generalization results.

VI. DISCUSSION

A. Evidence Summary

The study completes a traceable experimental workflow around the predefined evaluation protocol. The code runs the Language-Table environment, accepts multiple input forms, records GPT-5.5-based VLM responses and event triggers, evaluates raw controller variants and feedback sources, and exports sanitized per-run and aggregate evidence. The 2026-09-22 revision adds a fresh 16-run raw sweep, an 18-run live semantic suite, a nine-run target-image matrix, nine strict no-cache event controls, and six detector/tracker visual-feedback runs. A separate paired semantic-fault audit remains as controlled intervention evidence. The analysis preserves negative results rather than replacing them with only successful demonstrations.

The strongest positive evidence concerns oracle-feedback stabilization. Grounded original MPC makes the archived tasks stable under the 0.08 tolerance, but its 100% action-replacement rate means the raw planner does not control execution. The semantic-MPC suite achieves 18/18 selected-target control successes across a heterogeneous set of targets, input forms, seeds, and settings, but only 17/18 rows preserve requested-to-selected target identity and only the $n = 3$ seed subset directly measures seed variability. The target-image matrix provides 8/9 semantic matches and 8/9 joint active successes, with the failed selection retained. The semantic-fault audit shows 3/3 target repair and 2/3 true-target success under an explicitly forced query. These are useful diagnostic outcomes, but the 0/6 visual-feedback result shows that the oracle-to-vision replacement remains unresolved.

B. Evidence Boundary

The results do not prove that the original VLMPC paper has been reproduced at its reported performance. The fresh

TABLE X

REPEATED TARGET-IMAGE END-TO-END MATRIX. FINAL DISTANCE IS MEAN $\pm$ SAMPLE SD OVER ALL RUNS IN THE CASE. ACTIVE TRIALS EXCLUDE RUNS ALREADY WITHIN THE 0.08 THRESHOLD BEFORE THE FIRST ACTION; JOINT ACTIVE SUCCESS REQUIRES CORRECT SEMANTIC SELECTION AND ACTIVE CONTROL SUCCESS.

Target-image case	Runs	VLM semantic match	Pre-satisfied	Active joint success	Final distance	Interpretation
Red moon	3	3/3	0	3/3	0.0587 ± 0.0233	All three trials succeed
Blue cube	3	3/3	0	3/3	0.0623 ± 0.0249	All three trials succeed
Yellow pentagon	3	2/3	0	2/3	0.1640 ± 0.1512	Seed-42 semantic error; final distance 0.3386
All cases	9	8/9	0	8/9	0.0950 ± 0.0932	Small diagnostic matrix, not a benchmark

TABLE XI

OBSERVED SYMPTOMS IN THE FINAL AUDIT. THESE ARE NON-EXCLUSIVE OBSERVATIONS, NOT SINGLE-CAUSE PROOF.

Symptom	Evidence and interpretation
Large unmodified final distance	All 16 fresh raw-MPC runs end at 0.163–0.436, above 0.08; the reproduced video-prediction MPC chain is executable but not stable in our local setup.
Parameter changes do not rescue the run	Original ablation remains 0/8 across the tested sampling and cost settings; more extensive tuning may still be possible.
Visual-feedback gap	The detector/tracker feedback implementation remains 0/6 with final distance 0.3160 ± 0.1439 ; replacing oracle state with current visual estimates is insufficient.
Feedback sensitivity	Grounded diagnostics succeed after final-action state feedback is added; this does not prove raw MPC success.
Semantic mismatch	The old target-image heuristic often selects the wrong object; the repaired selector matches 4/4 audit cases and 8/9 matrix cases, with one yellow-pentagon error retained. The evidence is still small and should not be presented as broad robustness proof.

TABLE XII

ROBUSTNESS-EDGE AUDIT FOR TARGET DETECTION AND UNKNOWN-TARGET HANDLING.

Case	Status	Confidence
Original	ok	0.902
Occluded target	ok	0.632
Viewpoint proxy	ok	0.902
Brightness shift	ok	0.900
Purple pole	safe reject	–
Green moon	safe reject	–
Unknown target	safe reject	–

raw sweep fails in 0/16, and the visual-feedback controller fails in 0/6. The successful branches use privileged simulator-state feedback and are not directly comparable to the original video-prediction MPC results or published benchmarks. The semantic 18/18 count is conditional on the selected target; requested-to-selected matching and instruction-level joint success are both 17/18. The control result is also tolerance-dependent: semantic MPC falls from 18/18 at 0.08 to 3/18 at 0.05. Matched natural-event controls show no recovery advantage, while the paired forced-fault audit only establishes repair of a deliberately injected semantic error. Target-image grounding should therefore remain a small audited capability rather than a broad robustness claim.

C. Limitations and Future Work

The audit suggests three directions for stronger future work. First, the implemented detector/tracker visual-feedback controller should be improved rather than treated as an untested roadmap: its 0/6 result establishes the present boundary. The next evaluation should compare estimated target centroid, end-

effector pose, visibility, and goal distance against simulator state using centroid error, end-effector position error, goal-distance MAE, visibility F1, and temporal consistency. Visual-only and oracle feedback should then be compared under matched seeds using final distance, success at 0.05/0.08, recovery latency, and runtime per step. The implementation route is to export synchronized RGB/state pairs, calibrate the detector/tracker on held-out seeds and perturbations, and introduce confidence-gated fallback behind the same controller interface.

Second, event-triggered re-query should be tested over more instructions, perturbations, and no-requery/forced-requery/natural-event settings. The revision’s synthetic-fault audit establishes a measurable repair case but uses a forced query and therefore does not replace a natural-event causal study. A future study must induce a documented tracking or semantic failure, randomize the failure time, and compare enough matched seeds using final distance, recovery success, re-query count, recovery latency, and added VLM cost.

Third, target-image grounding should be evaluated with a larger set of image queries, cropped-object references, full-frame references, and manually verified semantic labels. These additions would convert the current diagnostic report into a stronger experimental paper without changing the interpretation of the current evidence.

D. Data and Code Availability

The public repository at <https://github.com/yixiaogithub66/vlmpc-language-table-reproduction-audit> and the accompanying v1.0.6 artifact include the manuscript source, authoritative 2026-09-22 per-run and aggregate CSVs, the portable robustness-edge CSV and inputs, exact method parameters, a reproducible figure script, source and runtime-asset manifests, and an automated consistency check. The artifact deliberately excludes credentials, private endpoint addresses, local absolute paths, virtual environments, raw service logs, and third-party model weights. Reported aggregates can be recomputed from the included portable files without cached VLM responses. End-to-end reruns additionally require independently obtained detector, tracker, and video-prediction checkpoints and a separately configured Responses-compatible VLM runtime, as documented in the reproduction guide.

VII. CONCLUSION

This paper presents a traceable audit of a VLMPC-style Language-Table project. The fresh unmodified video-

prediction MPC sweep is executable but fails in 0/16 runs, and the implemented detector/tracker visual-feedback controller fails in 0/6. Oracle-feedback diagnostics succeed under the $d_{\text{world}}^{\text{final}} \leq 0.08$ tolerance, including 18/18 selected-target semantic-MPC controls and 8/9 joint target-image trials, but these results rely on privileged simulator state; the semantic suite has 17/18 requested-to-selected matches and falls to 3/18 for control at 0.05. Fresh matched controls show that natural re-query executes without changing final distance, while forced semantic-fault intervention repairs 3/3 targets and reaches 2/3 true-target successes. The study is therefore a negative raw-reproduction audit plus bounded diagnostic evidence, not a successful unmodified VLMPC reproduction. The remaining technical problem is to close the measured oracle-to-vision gap under matched, multi-seed evaluation.

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